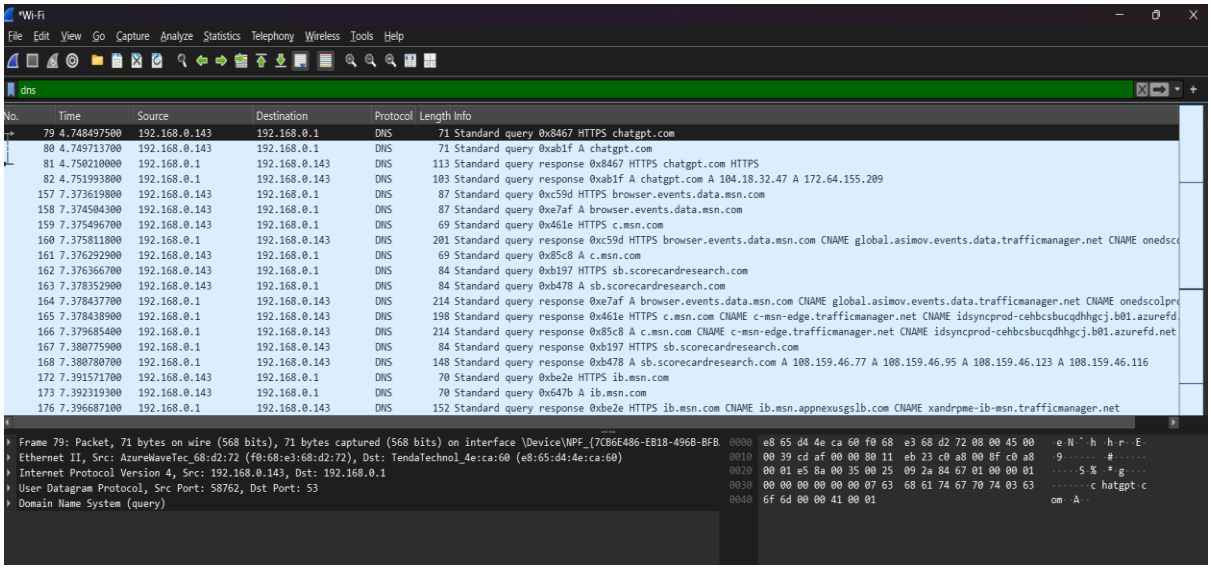
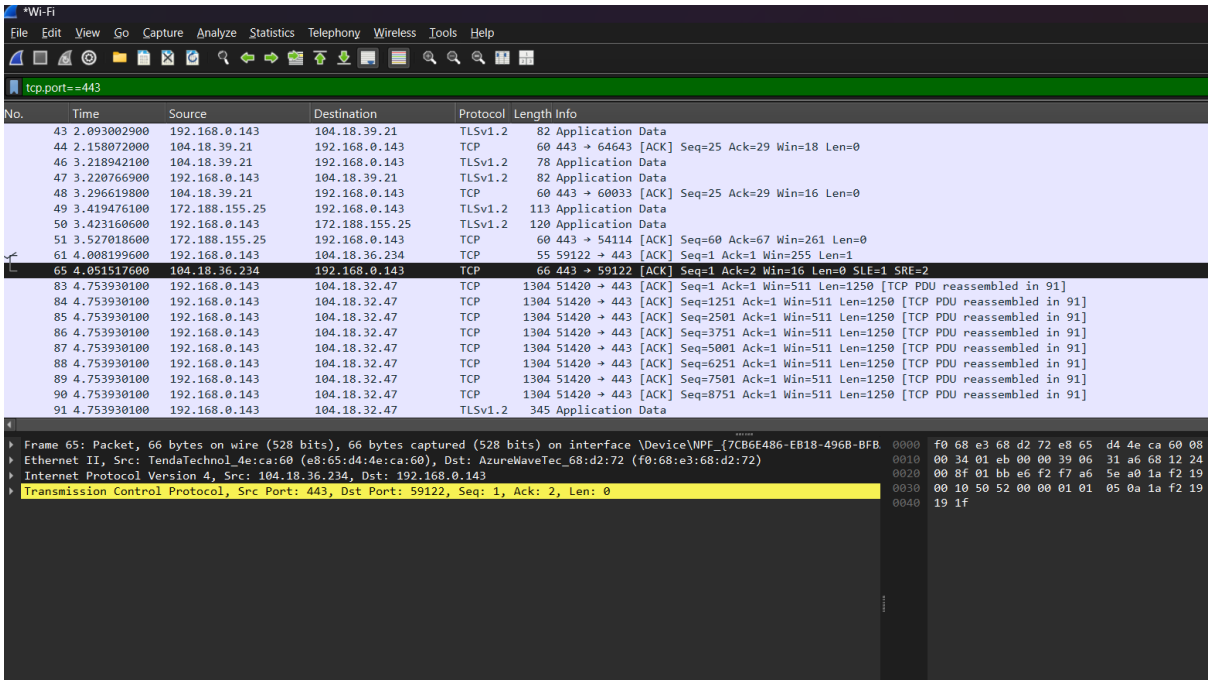
Task 1:

Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Answer:



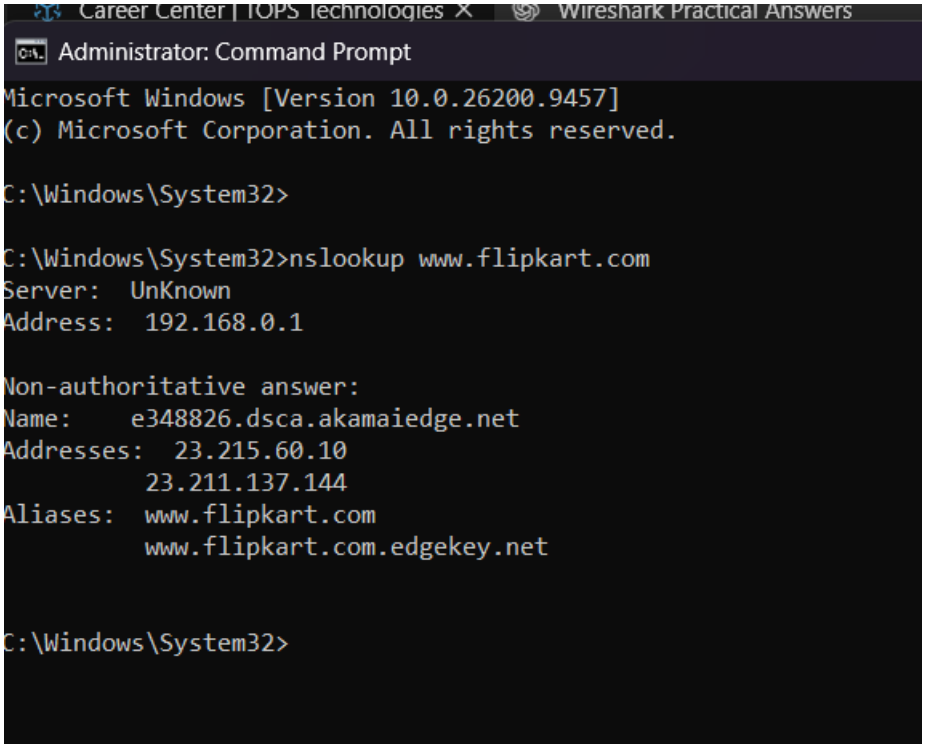


Wireshark was used to capture network traffic while accessing Spotify. DNS packets were used to resolve the domain name, and HTTPS traffic was used for secure communication with the website.

Task 2:

Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

Answer:



The nslookup command queries a DNS server to find the IP address associated with www.flipkart.com. This allows the computer to locate the server hosting the website.

Result: The DNS lookup was performed successfully, and the IP address returned by the DNS server was recorded.

Task 3:

Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

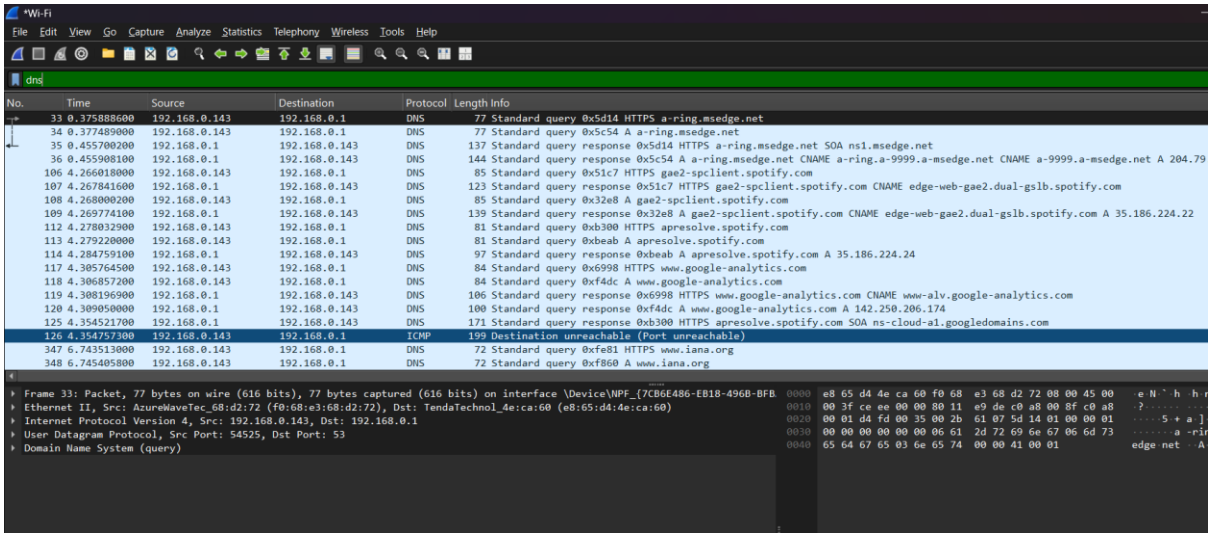
Answer:

No.	HTTP	HTTPS
1. Full form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
2. Security	Does not encrypt data by itself.	Encrypts data using TLS.
3. Encryption	Data is transmitted without HTTP-level encryption.	Data is encrypted during transmission.
4. Default port	Port 80	Port 443
5. URL format	Starts with http://	Starts with https://
6. Data privacy	Data can potentially be read if intercepted.	Encryption helps keep data private.
7. Authentication	Does not provide TLS-based server authentication.	Uses a digital certificate to authenticate the server.
8. Data integrity	Does not provide cryptographic integrity protection by itself.	TLS helps detect unauthorized modification of data.
9. Digital certificate	Does not require a TLS certificate.	Normally requires a valid TLS certificate for the website.
10. Banking and payments	Not suitable for sensitive transactions.	Essential for protecting sensitive transactions in transit.
11. SEO	HTTPS is a positive ranking signal for Google.	Can provide a ranking advantage compared with HTTP, all else being equal.
12. Browser indication	May be marked as “Not secure” by browsers.	Usually displays a secure connection indicator, depending on the browser.

Task 4:

Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Answer:



Task 5:

Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Answer:

Protocol selected: DHCP (Dynamic Host Configuration Protocol)

Scenario:

When I visit a cafe and connect my mobile phone to its Wi-Fi network to use Instagram, DHCP automatically assigns an IP address to my phone. It also provides other network settings, such as the subnet mask, default gateway, and DNS server.

DHCP uses a four-step process:

1. Discover: My phone searches for a DHCP server.
2. Offer: The DHCP server offers an available IP address.
3. Request: My phone requests that IP address.
4. Acknowledgment: The DHCP server confirms the assignment.

After receiving the network settings, my phone can communicate over the network and access Instagram.

Result: DHCP automatically provides the necessary network configuration, allowing devices to connect to Wi-Fi without manually entering an IP address.